

Homework I — Graphing Quadratics

Vertex Form, Completing the Square & Putting It Together

Name: _____ Date: _____

Instructions

Use a **pencil** and **show your work**. Try each problem yourself before checking anything. You are allowed to use AI for help, but always try the problem yourself first.

Reading 0 — Vertex Form, at a Glance

Every quadratic can be written in **vertex form**

$$y = a(x - h)^2 + k.$$

The three parameters each have one job:

- a controls how “steep” the parabola is, h shifts the parabola left/right, and k shifts it up/down. The vertex sits at (h, k) , and the axis of symmetry is the vertical line $x = h$.
- **Be careful of the sign of h .** The form has a *minus*: $(x - h)$. So $y = (x - 3)^2$ has its vertex at $(3, 0)$, while $y = (x + 1)^2 = (x - (-1))^2$ has its vertex at $(-1, 0)$.

Section 1 — Graphing from Vertex Form (and backwards)

Reading — a reliable way to graph

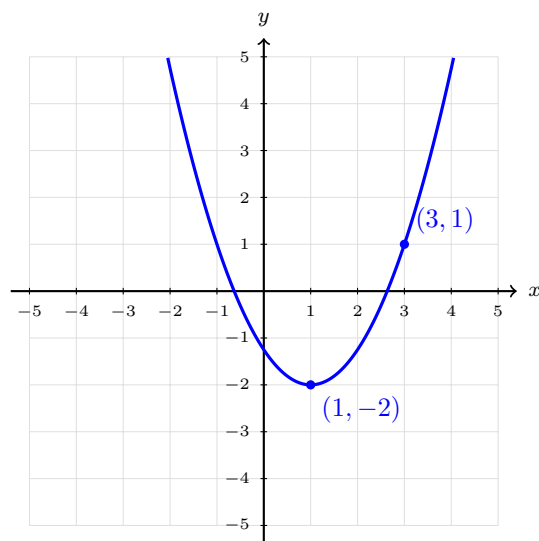
Given $y = a(x - h)^2 + k$, graph it in four steps:

1. Plot the vertex (h, k) .
2. Use a to decide which way it opens (up or down) and how steep it is.

Going backward — from a graph to an equation

To read an equation off a graph: the vertex gives you h and k immediately. You can then look at how the height changes to find a . However, for not-so-obvious a 's, such as $a = 3/4$, this will be hard to see by eye. So a more reliable method that will always work is to pick one more point (x, y) that the curve clearly passes through, substitute it in, and solve for a .

Example



The parabola above has its vertex at $(1, -2)$, so $h = 1$ and $k = -2$:

$$y = a(x - 1)^2 - 2.$$

The stretch factor a is not something you can read off by eye here — it turns out to be a fraction. So use one more point the curve clearly passes through, say $(3, 1)$, and substitute

it for x and y :

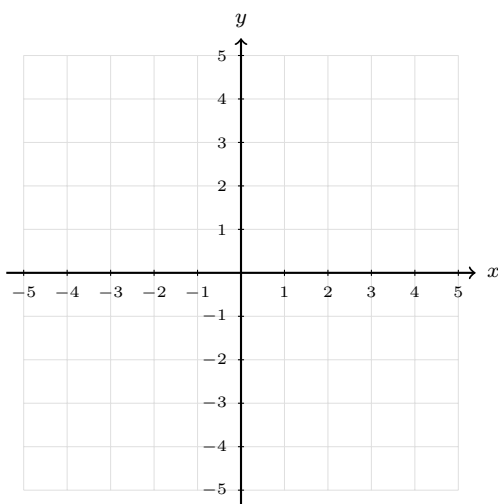
$$1 = a(3 - 1)^2 - 2 \implies 1 = 4a - 2 \implies 4a = 3 \implies a = \frac{3}{4}.$$

So the equation is $y = \frac{3}{4}(x - 1)^2 - 2$. (Why does this method work?)

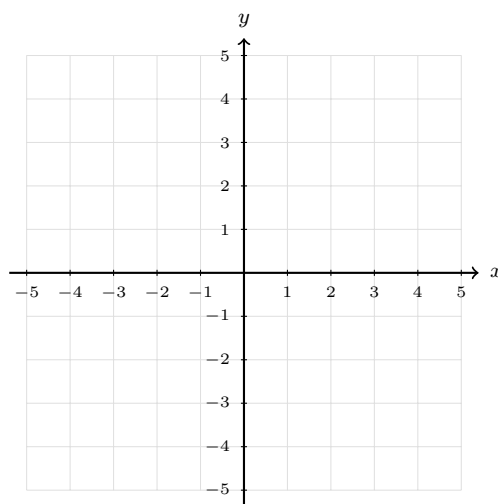
Problems 1–10 — graph each quadratic

Mark the vertex, then plot enough points to show the shape.

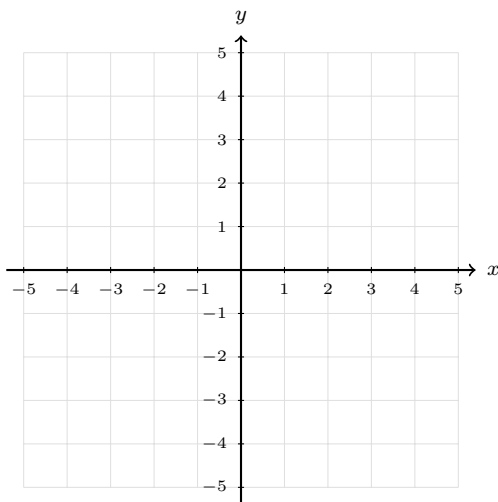
1. $y = (x - 2)^2 - 3$



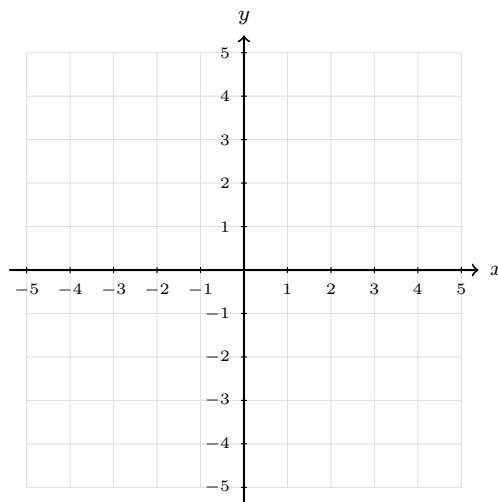
2. $y = (x + 1)^2 - 4$



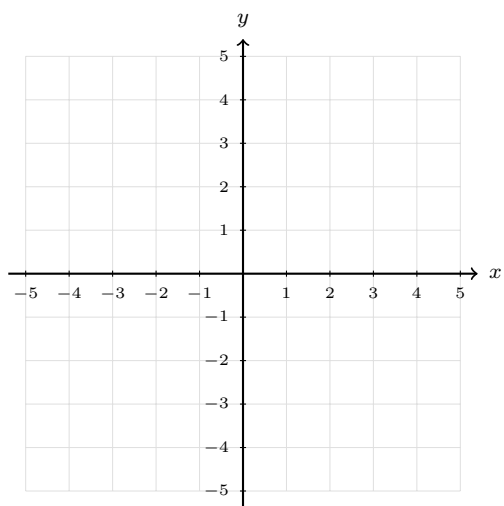
3. $y = -(x - 1)^2 + 3$



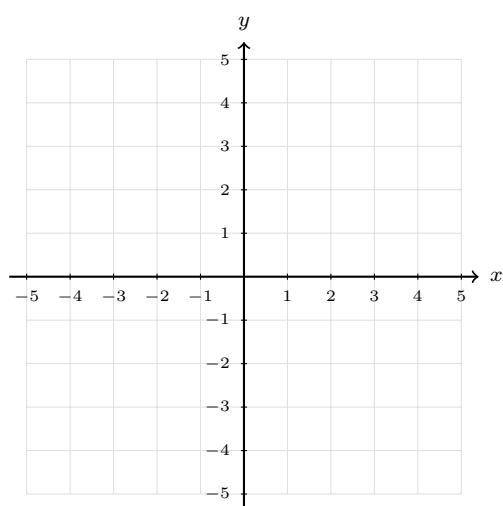
4. $y = -(x + 2)^2 + 2$



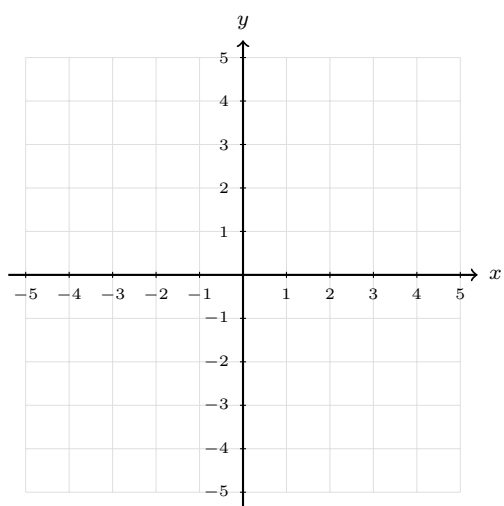
5. $y = 2(x - 1)^2 - 2$



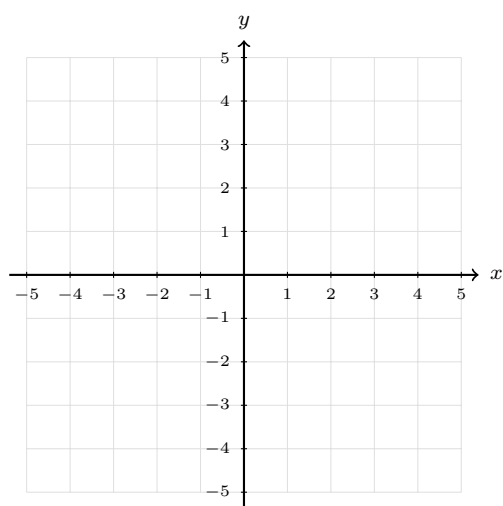
6. $y = \frac{1}{2}(x + 2)^2 - 3$



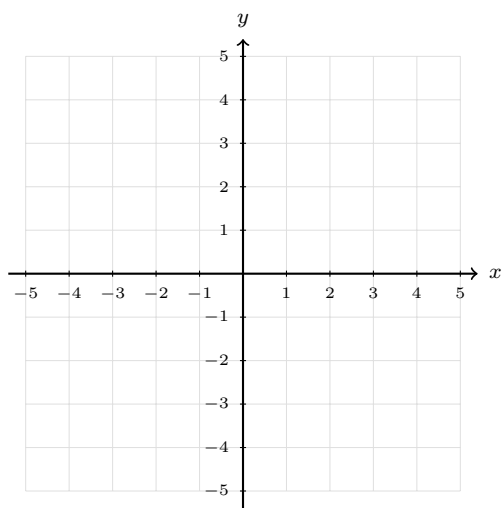
7. $y = -2(x - 2)^2 + 4$



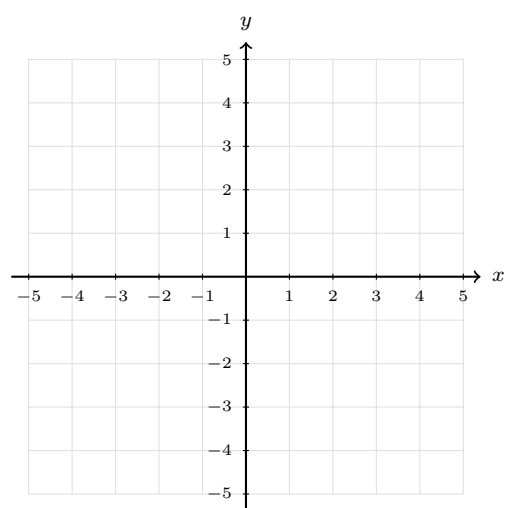
8. $y = (x - 3)^2 - 1$



9. $y = -(x + 3)^2 + 4$



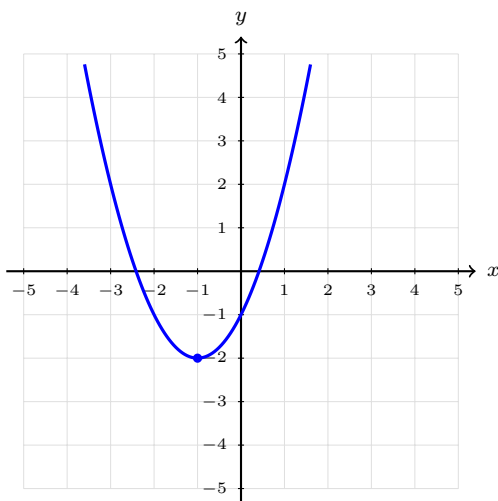
10. $y = \frac{1}{2}(x - 1)^2 - 2$



Problems 11–15 — write the equation of each graph

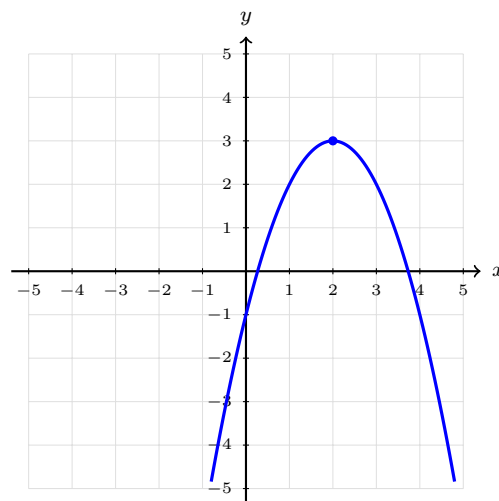
Each graph is a parabola in the form $y = a(x - h)^2 + k$. Read off the vertex, use one more point (if necessary), and write the equation.

11.



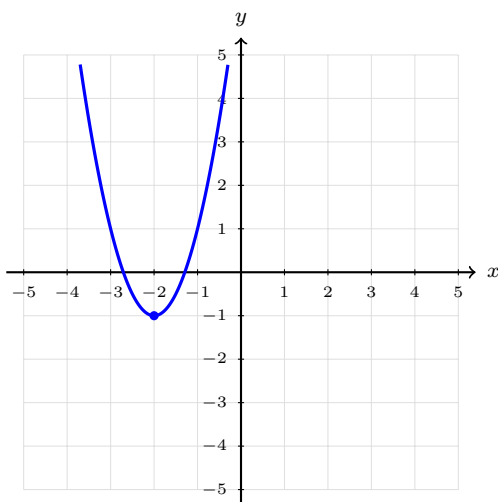
Equation: _____

12.



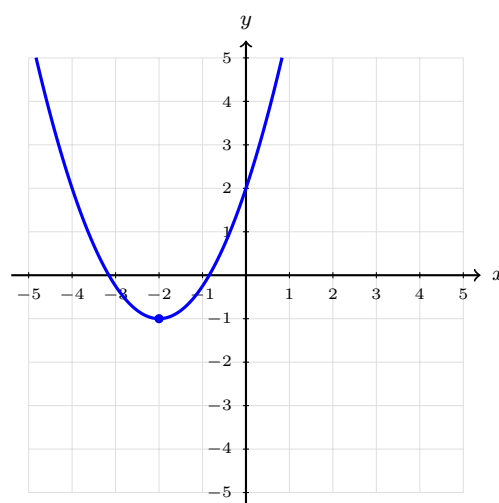
Equation: _____

13.



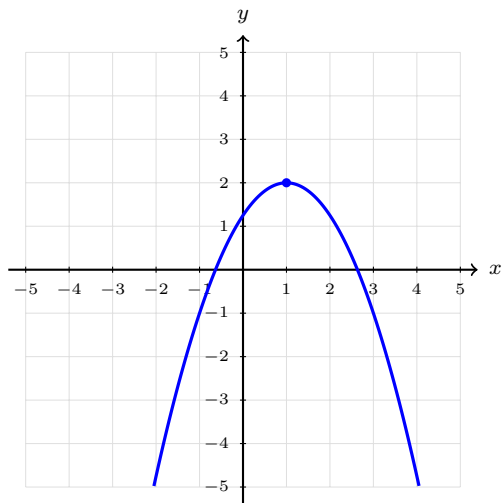
Equation: _____

14. (Notice a is not so visually obvious)



Equation: _____

15.



Equation: _____

Section 2 — Completing the Square

Reading — turning $ax^2 + bx + c$ into vertex form

Worked example ($a = 1$). Rewrite $x^2 + 6x + 1$ in vertex form.

$$\begin{aligned}x^2 + 6x + 1 &= \left(x^2 + 6x + \left(\frac{6}{2}\right)^2\right) - 9 + 1 && \text{add and subtract } (b/2)^2 \\&= (x^2 + 6x + 9) - 8 \\&= (x + 3)^2 - 8. && \text{fold the perfect square}\end{aligned}$$

The key idea here is: take half the middle coefficient, **square it**, and add-and-subtract it so you can fold a perfect square without changing the value.

Worked example ($a \neq 1$). For $2x^2 - 12x + 5$, first **factor a out of the x -terms**:

$$2x^2 - 12x + 5 = 2(x^2 - 6x) + 5 = 2(x^2 - 6x + 9 - 9) + 5 = 2((x - 3)^2 - 9) + 5 = 2(x - 3)^2 - 13.$$

Problems 16–21 — complete the square

Write each in the form $a(x - h)^2 + k$. Show your steps in the space below each problem.

16. $x^2 + 6x + 5$

17. $x^2 - 4x + 1$

18. $x^2 + 3x + 1$

19. $2x^2 - 12x + 7$

20. $3x^2 + 12x + 5$

21. $-x^2 + 4x + 1$

Section 3 — Putting It All Together

Reading — you cannot “eyeball” a parabola from $ax^2 + bx + c$

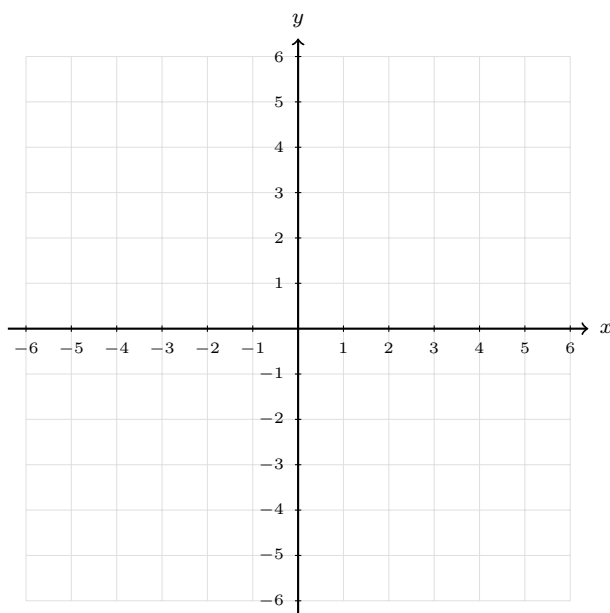
Looking at a raw quadratic like $x^2 - 6x + 11$, you *cannot* tell where its vertex is or whether it crosses the x -axis — your intuition has almost nothing to grab onto. Completing the square is what reveals the picture. Once it is in vertex form you can read the vertex, the direction, and the intercepts directly.

Problems 22–25

For each: (i) complete the square to get vertex form (ii) graph it.

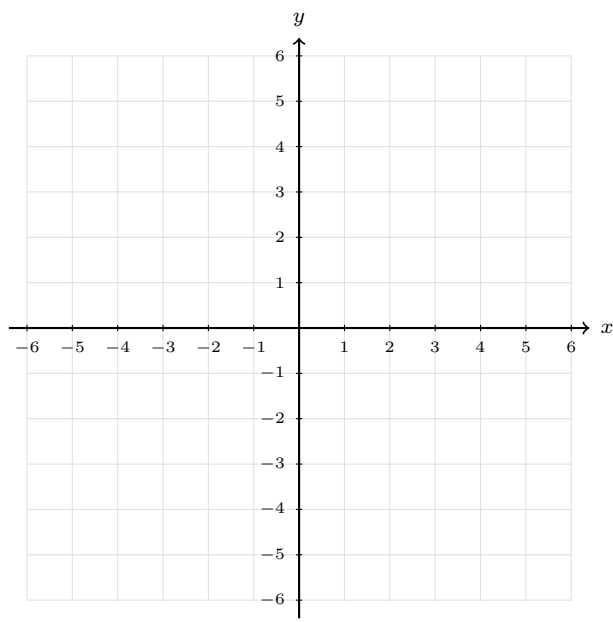
22. $y = x^2 - 6x + 11$

Vertex form: _____



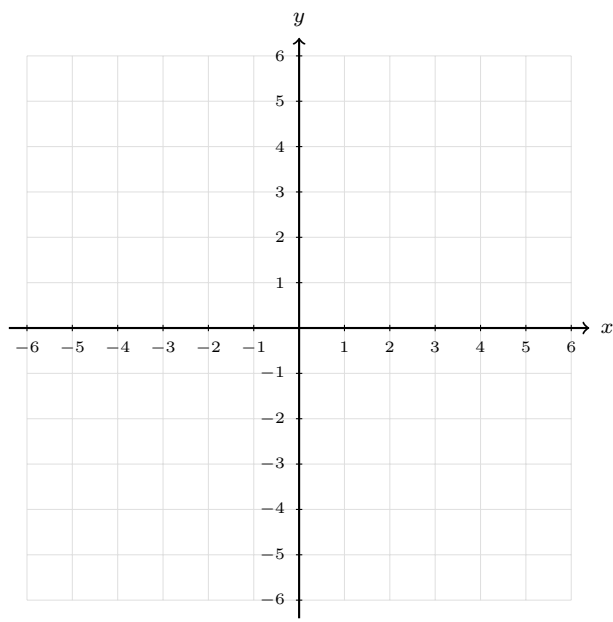
23. $y = x^2 + 4x + 1$

Vertex form: _____



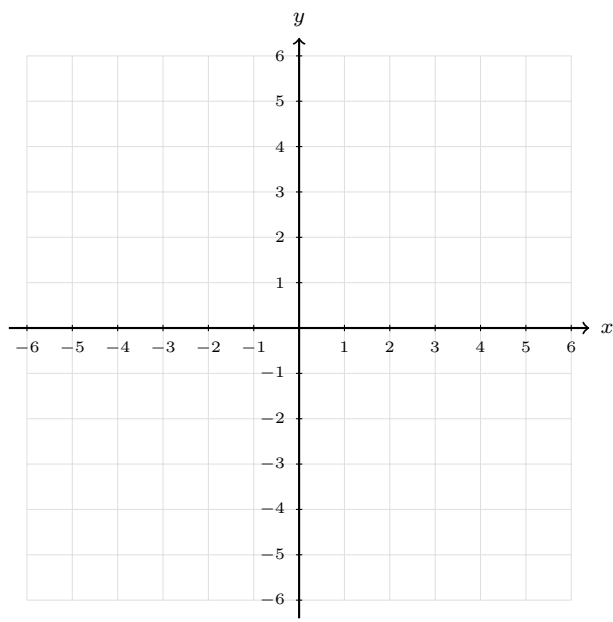
24. $y = -x^2 + 2x + 3$

Vertex form: _____



25. $y = 2x^2 - 8x + 5$

Vertex form: _____



★ Challenge Problems

26. ★ The biggest pen. A rancher has 40 meters of fencing to make a rectangular pen, using a straight wall as one side (so the wall side needs no fence). All 40 meters go into the other three sides (which you are deciding the length of them). Let w be the length of each of the two sides that stick out from the wall.

(a) As defined, the two sides sticking out from the wall each have length w , and the remaining fence forms the side parallel to the wall. Using the fact that the three fenced sides total 40 meters, write the length of the side parallel to the wall in terms of w . Then, using $\text{area} = \text{length} \times \text{width}$, write a function $A(w)$ for the enclosed area. (*Sketching the pen first will help.*)

(b) Expand $A(w)$ and complete the square (or use the vertex formula) to find the width w that gives the **maximum** area, and state that maximum area. Explain why the vertex answers this question.

27. ★ Where the quadratic formula comes from. Complete the square on the general quadratic to *derive* the quadratic formula. Start from

$$ax^2 + bx + c = 0 \quad (a \neq 0),$$

and follow the same steps you used all homework: factor a out of the first two terms, complete the square, then solve for x . You should arrive at

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}.$$

Instructor Answer Key

Section 1 (1–10), vertex form $\rightarrow$ vertex & direction:

- 1. $(2, -3)$ up 2. $(-1, -4)$ up 3. $(1, 3)$ down 4. $(-2, 2)$ down
- 5. $(1, -2)$ up, narrow 6. $(-2, -3)$ up, wide 7. $(2, 4)$ down, narrow 8. $(3, -1)$ up
- 9. $(-3, 4)$ down 10. $(1, -2)$ up, wide

Section 1 (11–15), read-the-graph equations:

- 11. $y = (x + 1)^2 - 2$ 12. $y = -(x - 2)^2 + 3$ 13. $y = 2(x + 2)^2 - 1$
- 14. $y = \frac{3}{4}(x + 2)^2 - 1$ 15. $y = -\frac{3}{4}(x - 1)^2 + 2$

Section 2 (16–21), completed square:

- 16. $(x + 3)^2 - 4$ 17. $(x - 2)^2 - 3$ 18. $(x + \frac{3}{2})^2 - \frac{5}{4}$
- 19. $2(x - 3)^2 - 11$ 20. $3(x + 2)^2 - 7$ 21. $-(x - 2)^2 + 5$

Section 3 (22–25):

- 22. $(x - 3)^2 + 2$; vertex $(3, 2)$; **0** x -intercepts (the “gotcha” — never touches the axis).
- 23. $(x + 2)^2 - 3$; vertex $(-2, -3)$; **2** x -intercepts $(x = -2 \pm \sqrt{3})$.
- 24. $-(x - 1)^2 + 4$; vertex $(1, 4)$, opens down; **2** x -intercepts $(x = -1, 3)$.
- 25. $2(x - 2)^2 - 3$; vertex $(2, -3)$, narrow; **2** x -intercepts.

Challenges:

- 26. $A(w) = 40w - 2w^2 = -2(w - 10)^2 + 200$; max area **200** m^2 at $w = 10$ (so the pen is 10×20).
- 27. $ax^2 + bx + c = 0 \Rightarrow a(x + \frac{b}{2a})^2 = \frac{b^2}{4a} - c \Rightarrow (x + \frac{b}{2a})^2 = \frac{b^2 - 4ac}{4a^2} \Rightarrow x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.